Status	Finished
Started	Tuesday, 20 October 2026, 5:27 PM
Completed	Tuesday, 20 October 2026, 5:31 PM
Duration	3 mins 49 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list is very small
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **2**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search works on unsorted lists
c. Binary search can find multiple instances of the target element
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **3**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓
b. When the list contains duplicate elements
c. When the list is sorted
d. When the search algorithm finishes

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. Must use a sorted array

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search backtracks to find duplicate elements
c. The search starts over from the beginning
d. The search stops immediately ✓

Question **6**
Correct
Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Dividing the list in half repeatedly
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **7**
Correct
Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Linear search ✓
b. Binary search
c. Hashing
d. Dynamic search

Question **8**
Correct
Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The search continues from the beginning
d. The list is sorted

Question **9**
Correct
Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(n log n)
c. O(1) ✓
d. O👎

Question **10**
Correct
Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 90 and 99 ✓
c. 89 and 99
d. 90 and 94

Question **11**
Correct
Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n ✓
b. log n
c. n/2
d. n-1

Question **12**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Selection

c. Bubble ✓

d. Insertion

Question **13**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Deleting elements from a list

b. Inserting elements into a list

c. Determining whether an element is present in a list ✓

d. Sorting elements in a list

Question **14**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing

b. Jump search

c. Binary search

d. Linear search ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓

b. At the beginning of the list

c. In the entire list

d. In the right sublist